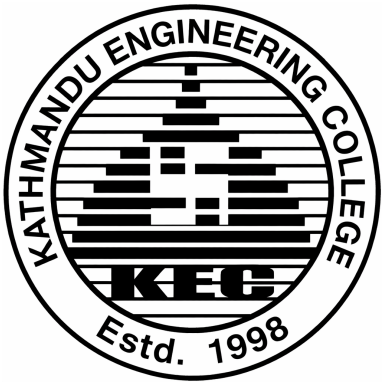
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Project Report on

SPARK: Explainable Edge AI for Kinetic Pattern Recognition and Distress Signaling

Final-Defense Thesis Draft

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Defense date: awaiting institutional confirmation

Report Status and Authorship

This editable thesis draft records implementation and available evidence as inspected on October 14, 2026. It is not certification of a completed clinical or hardware evaluation. The project title, student names and rolls, supervisor, coordinator, and department leadership are retained from the KEC proposal. Result-dependent sections remain explicitly pending until hardware and approved cohort evidence are available.

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Primary completion objective: human fall detection with a validated wearable logger, approved cohort protocol, and integrated evaluation. Robot fall monitoring is an open research extension with separate data, adaptation, and validation requirements.

Evidence labels distinguish **current inspection**, **archived measured results**, **historical reported results**, **estimates**, and **planned work**. Reading an artifact confirms its existence and contents; it does not independently reproduce its reported experiment.

Approval and Declaration Forms — Draft

Unsigned preparation copy. These fields preserve institutional roles for later completion. They do not certify review, submission, approval, originality sign-off, or successful defense. The project title is *SPARK: Explainable Edge AI for Kinetic Pattern Recognition and Distress Signaling*.

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Abstract

SPARK currently provides offline human-fall model artifacts and tested gateway software, while its physical wearable pipeline remains incomplete. The intended system combines a threshold gate and quantized convolutional classifier on an ESP32-S3 with local incident storage, explanation, and PDF reporting on a laptop gateway. A display client consumes gateway records rather than performing BLE reception or explanation on the phone.

The existing model preparation uses 100 Hz, two-second windows of 200 samples and six channels. Archived August evaluation reports give 87.81% floating-point sensitivity and 0.9185 AUC-ROC on 6,712 held-out windows; the corresponding INT8 report gives 87.43% sensitivity and 0.8717 AUC-ROC. A separate October Colab log reports 89.81% sensitivity and 0.9272 AUC-ROC, with unresolved threshold provenance. These are archived experimental records inspected for this report, not rerun evaluations or physical wearable results.

Driver, interpreter, and transport stubs prevent a claim of integrated MCU operation. Hardware timing, battery endurance, and approved Nepal cohort results remain unavailable. This thesis draft develops the system requirements, mathematical formulation, implementation audit, result analysis, and future evaluation protocol using established evidence. Hardware- and cohort-dependent sections are marked pending. Dependency gates replace calendar schedules. Human fall detection remains the completion objective; a planned forty-episode MuJoCo study appears only as an exploratory robotics appendix.

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List of Abbreviations

ADL	Activities of Daily Living
API	Application Programming Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
AVX-VNNI	Vector Neural Network Instructions (AVX)
BEI	Bachelor of Electronics, Communication and Information Engineering
BLE	Bluetooth Low Energy
CNN	Convolutional Neural Network
CORS	Cross-Origin Resource Sharing
DFM	Design for Manufacturability
ESP32-S3	Espressif Systems ESP32-S3 Dual-Core Xtensa LX7 SoC
GATT	Generic Attribute Profile (Bluetooth LE)
IMU	Inertial Measurement Unit
INT8	8-bit Integer Quantization
IOE	Institute of Engineering
ISO	International Organization for Standardization
ISR	Interrupt Service Routine
KEC	Kathmandu Engineering College
MPU6050	InvenSense MPU-6050 6-axis IMU
NPU	Neural Processing Unit
OpenVINO	Open Visual Inference and Neural Network Optimization
SHAP	SHapley Additive exPlanations
SPARK	Signal Pattern Analysis & Real-time Kinetics
TFLite	TensorFlow Lite
TPU	Thermoplastic Polyurethane
WHO	World Health Organization
XAI	Explainable Artificial Intelligence

Chapter 1: Introduction

1.1 Problem and motivation

SPARK investigates whether a local wearable-and-gateway system can detect human falls and preserve inspectable incident records. The engineering problem includes false alarms during daily activities, incomplete sensor data, limited wearable resources, and the gap between an offline classifier and an integrated device. A classification score alone cannot establish that a complete fall event is detected or delivered to a caregiver.

The intended wearable observes six inertial channels and performs gated classification. The gateway stores event records and prepares explanations and reports locally. This architecture motivates experiments on computation, detection quality, and evidence traceability; it does not establish clinical effectiveness or superiority over commercial systems.

1.2 Potential applications

If the integrated design is later validated, a wearable alert could support follow-up for older adults or other people at fall risk in settings where a fixed room sensor is unavailable. A local gateway could preserve event records for review and a display client could present alerts to an authorized caregiver. These are intended use cases only. The present software does not contact emergency services, continuously track location, or provide a medical diagnosis. The proposal's broader aquatic and robot applications are deferred and are not evidence-backed extensions of the human fall model.

1.3 Objectives and research questions

The primary objective is to complete and evaluate human fall detection. The implementation must recover an explicit sampling, units, and model-input contract; replace hardware interfaces with real acquisition and inference; and evaluate event detection after approval of the cohort protocol.

1. Can the implemented gate retain fall events while limiting classifier invocations during non-fall exposure?
2. What sensitivity, false-alarm rate, timing, and quantization loss occur under a frozen subject/episode holdout?
3. Can the wearable, gateway, and display preserve event identity and provenance through the reporting pipeline?

4. What limitations arise when transferring waist-worn public data to the proposed dorsal-wrist placement?

1.4 Scope and limitations

The evidence presently concerns SisFall-derived offline windows, stored model outputs, source inspection, and software tests. The proposed Nepal cohort is not a collected dataset. A protocol draft does not constitute institutional approval. Young volunteer simulations would not, by themselves, validate performance in frail older adults or natural falls.

Robotic sensing, emergency-stop interfaces, active fall mitigation, aquatic monitoring, and physical robot experiments are out of scope for the human completion milestone. Appendix B defines an unexecuted experiment rather than an implemented capability. No first-in-literature or production-readiness claim is made.

1.5 Evidence conventions

Current inspection means directly observed file contents, artifact identity, array metadata, or checks executed for this preparation. **Archived measured results** means numerical outputs retained from an earlier experiment and not independently rerun here. **Historical reported results** means tracker or document narratives without a corresponding raw run record inspected in this preparation. **Estimates** require assumptions; **planned work** requires completion evidence before becoming a result.

The appendix evidence register maps claims to repository paths and the local snapshots in this report folder. Source existence does not prove physical deployment. Result-dependent sections remain pending; defense dates and procurement dates are not inferred from old schedules.

Chapter 2: Background and Related Work

2.1 Fall detection and application context

Falls are an important source of injury and loss of independence among older adults. A wearable detector is intended to identify a suspected event when a person is away from fixed cameras or room sensors, preserve a traceable event record, and support a caregiver's follow-up. Such a detector is an assistive alerting tool; it does not diagnose injury, replace emergency services, or establish that a person is safe when no alarm occurs. WHO guidance motivates the broad fall-prevention context, and Nepal's census-based population projections provide demographic planning context; neither source measures local fall prevalence or demand for SPARK [1, 2].

The research literature spans threshold rules, feature-based classifiers, temporal neural networks, edge inference, and systems that send candidate signals to a server. Each choice affects more than classification accuracy: sensor placement, event definitions, false alarms over ordinary activity, compute and radio duty, notification delay, usability, and evaluation population all matter. SisFall is a public benchmark for reproducible development, but its waist-mounted signals and largely acted falls do not validate a dorsal-wrist system or natural falls [3].

2.2 Comparative literature review

Threshold methods are computationally simple and interpretable. Kangas et al. compared low-complexity algorithms at waist, wrist, and head placements; their results emphasize that the best site and rule depend on placement, and that the wrist signal poses a particular challenge [4]. Bourke et al. evaluated tri-axial thresholds using simulated falls and older adults' daily activities, illustrating both the value of trunk signals and the limits of transferring a threshold across sensor positions [5]. These studies motivate a gate as a candidate filter, not the assumption that a fixed gate preserves every fall.

Learned models can represent temporal patterns that a single magnitude rule cannot. SisFall provides 15 fall activities and 19 activities of daily living from 38 participants, with inertial sensors; it is a useful public source for offline model development, subject-grouped evaluation, and comparison of preprocessing choices [3]. CNN and recurrent approaches represent temporal structure differently. Nho et al.'s cited title and the listed Wang et al. citation could not be matched to authoritative primary records using the supplied metadata; this report therefore does not repeat their performance claims or use them to rank architectures. Those entries remain visible in the proposal-reference audit.

Embedded inference literature establishes a feasible runtime route, not a guarantee for this board and model. TensorFlow Lite Micro describes an interpreter and evaluation for constrained embedded systems; actual RAM, timing, operator coverage, and whole-firmware size still require target measurements [6]. Xu et al. provide a directly relevant hybrid example: their wearable threshold stage forwards candidate acceleration windows to a server-side CNN, so the classifier is not fully local to the wearable [7]. Yu et al. report a TinyCNN wearable prototype and an Arduino Nano BLE 33 Sense implementation. Its published record is in Biomedical Signal Processing and Control, volume 86, article 105325 (2023), not the Measurement citation given in the proposal [8]. Hu et al. report MicroFallNet on UMAFall and FallAllD, including an ESP32 implementation; the correct record is Pervasive and Mobile Computing, volume 109, article 102046 (2025), not volume 107/article 101980 [9].

Other system designs broaden the comparison beyond classifier topology. Tseng et al. describe a finite-state wearable detector with GPS, NB-IoT, and notifications, demonstrating a location-aware communication path distinct from SPARK's local gateway design [10]. Bhattacharyya et al. describe a CNN-kNN fall detector on a RISC-V system implemented on FPGA; this is an accelerator-oriented hardware point of comparison, not evidence that a low-cost MCU has equivalent resources [11]. Baños et al.'s mHealthDroid is relevant to mobile-health framework context, but SPARK's phone client is deliberately display-only and does not adopt its broader sensor integration architecture [12].

Explainability terminology also constrains the comparison. SHAP defines additive feature attributions grounded in Shapley values [13]. The SPARK gateway's input-times-gradient calculation is a different attribution method, while the live v1 event's peak-feature display is a heuristic summary because the packet does not contain a waveform. Neither should be labeled SHAP. Yao et al. study ECG heartbeat interpretation using CNN and GRU; that biomedical time-series example supplies context but is not fall-detection evidence [14]. Liu et al.'s IEEE conference paper concerns explainable wearable human-activity recognition rather than fall-event explanation [15]. Kundu et al.'s review is an arXiv preprint submitted in 2026 (arXiv:2604.09799), rather than a 2025 peer-reviewed article as the proposal bibliography says [16].

2.3 Hardware and software theory

An inertial measurement unit combines accelerometer and angular-rate measurements. A three-axis accelerometer observes specific force in its sensor frame; a gyroscope observes angular velocity. Orientation, gravity separation, scale factors, axis alignment, and units must be considered before thresholds or models trained on one representation are applied to another. The current preprocessing selection retains raw SisFall sensor counts, whereas the event protocol describes peak values in g and degrees per second. The exact transform

Table 2.1: Comparison of verified and directly relevant approaches

Approach	Evidence in cited work	Design implication	SPARK status and boundary
Threshold rules	Low-complexity inertial algorithms; placement matters [4, 5]	Measure gate sensitivity and false alarms at the intended placement	Gate code exists; its 2.5 g / greater-than-300 ms behavior awaits event-level evaluation
Public inertial benchmark	SisFall contains falls and daily activities from 38 people [3]	Use grouped splits and disclose sensor placement and acted-fall limits	Offline 100 Hz, 200-sample, six-channel windows; wrist transfer remains unvalidated
Embedded runtime	TFLite Micro targets constrained embedded inference [6]	Measure linked target memory, operators, and latency	Local model bytes are known; MCU inference remains a stub
Hybrid gate plus CNN	Xu et al. send candidate windows to a server CNN [7]	Report compute location and communication dependency explicitly	SPARK intends local inference, but has not demonstrated it on hardware
Wearable edge CNN	TinyCNN and MicroFallNet report microcontroller implementations [8, 9]	Treat published results as feasibility context, not a head-to-head comparison	Different datasets, placements, models, and protocols prevent direct score comparison
Location and FPGA systems	Tseng et al. include localization/notification; Vigil uses FPGA-oriented RISC-V hardware [10, 11]	Compare system goals and hardware classes, not only accuracy	Context only; neither architecture is the SPARK implementation
Explanation methods	SHAP has Shapley-value semantics; HAR/ECG work covers different tasks [13–16]	Name the actual attribution method and input used	Gradient-times-input and peak-feature fallback are distinct; explanation validity is pending

linking those representations is unresolved and must be specified before firmware integration.

The intended embedded design uses an ESP32-S3 and MPU6050. Sensor acquisition, timestamping, buffering, preprocessing, gate evaluation, model invocation, and event delivery form a real-time chain. A gateway receives and stores events, generates local reports, and offers a display API. The phone is a display client. The architecture is staged to make failures inspectable; the current sensor reads, TFLite invocation, transport writes, and logger loop are not complete physical implementations.

For six input channels, a temporal convolution computes learned local filters over adjacent samples. Pooling reduces temporal resolution, global average pooling summarizes activations, and a dense output layer produces two class scores. INT8 quantization represents a real value approximately as $x = s(q - z)$ using model-specific scale s and zero point z. This reduces weight storage but introduces quantization error and does not account for runtime buffers or the rest of the firmware. These are engineering mechanisms, not performance guarantees.

2.4 Interpretation and study position

SPARK's defensible present contribution is an inspectable software pipeline, offline model artifacts, and an evaluation design that separates archived results from future integrated measurements. The literature supports testing threshold filtering, quantized embedded inference, event traceability, and local explanation as design questions. It does not establish novelty, clinical suitability, power savings, or superiority for this system. The exact source records and claim-level disposition for all twenty proposal references are documented in `docs/thesis_work/PROPOSAL_COVERAGE_AND_REFERENCE_AUDIT.md`.

Chapter 3: Architecture and Methodology

3.1 System allocation

The intended node uses an ESP32-S3 and MPU6050. The laptop gateway owns transport reception, local storage, explanation, and PDF generation. The mobile/display layer reads the gateway API. Figure 3.1 separates the intended wearable from implemented gateway modules.

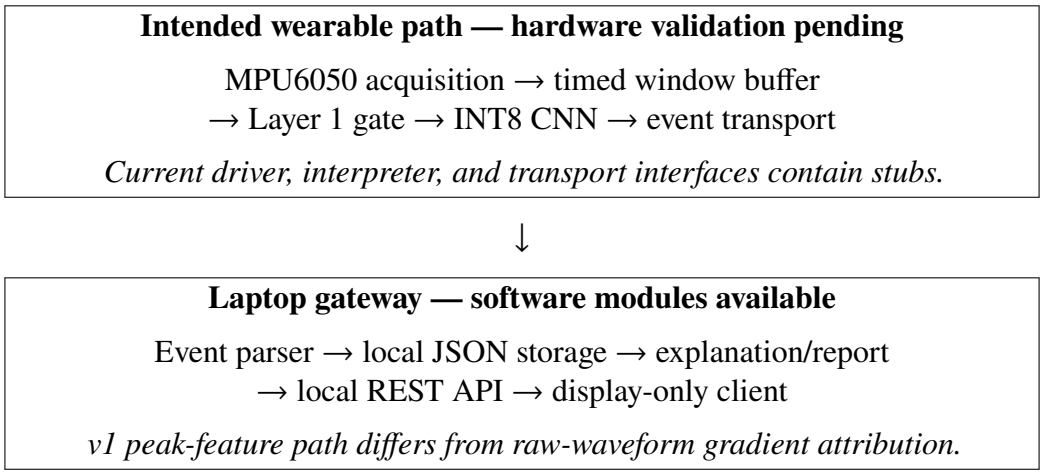


Figure 3.1: Intended architecture with current implementation boundaries. The figure is a design map, not evidence of integrated device operation.

3.2 Sampling and input contract

Let f_s be sampling frequency and N the number of samples. The nominal window duration is $T = N/f_s$. Existing preprocessing sets $f_s = 100\,\text{Hz}$ and $N = 200$, hence $T = 2\,\text{s}$. Model arrays have six channels ordered as acceleration then angular-rate axes. The planned logger targets 200 Hz acquisition; it is not evidence that the model was trained at that rate.

A 200-sample stream at 200 Hz has one second of coverage. A three-second 200 Hz input requires 600 samples. Consequently, the historical combination "200 Hz, three seconds, 200×6" is rejected. This report retains the executable model convention and requires a versioned resampling adapter for any higher-rate logger. It does not silently alter the trained model.

The source preparation retains sensor counts in the selected columns; it does not establish a calibrated physical-unit conversion. Before connecting firmware, reconcile accelerometer/gyro scaling, axis signs, normalization, interpolation, anti-aliasing, and quantization. Timestamp gaps, clipping, and non-finite input must be explicit rejection or quality flags.

3.3 Gate and classifier

The acceleration magnitude is

$$a_m = \sqrt{a_x^2 + a_y^2 + a_z^2}.$$

Current host gate code triggers when magnitude is at least 2.5 g and the continuous above-threshold duration exceeds 300 ms. This differs from older prose describing a shorter-than-300 ms spike. The implemented behavior must be evaluated against annotated falls before either condition is accepted as the physical design. No pre-impact lead time is established.

The CNN architecture uses two Conv1D stages, pooling, global average pooling, a dense hidden stage, and two-class softmax output. Class 1 represents FALL and class 0 represents non-fall. For a frozen decision threshold τ, the binary decision is

$$\hat{y} = \begin{cases} 1 & \text{if } P(\text{FALL}) \geq \tau \quad \text{(fall)}, \\ 0 & \text{otherwise} \quad \text{(non-fall)}. \end{cases}$$

Threshold tuning belongs on validation data; test labels must not select τ.

Full-integer quantization maps a float input to $q = \text{clip}(\text{round}(x/s) + z)$ using model-specific scale s and zero point z. The loaded artifact must supply these values; they cannot be borrowed across model generations. Quantized output dequantization precedes threshold comparison.

3.4 Holdout and evaluation design

The training source uses grouped holdouts and asserts subject-set disjointness. Group-based fractions need not yield identical fractions of windows, and the helper is not repeated cross-validation. The first source split reserves a test group fraction of 0.2; the second split uses a validation fraction of $0.1/0.9$ within the retained groups. Do not describe this code as a guaranteed 80/10/10 window split. The October notebook run must be traced separately from the CLI source.

Retain exact subject/episode IDs, class counts, seeds, preprocessing revision, and hashes for each split. Trial-level labels inherited by every window can include pre/post-fall periods; event-level ground truth must eventually resolve onset and contact rather than treating every such window as a separate fall.

3.5 Event and explanation contract

Wire Format v1 records event identity, device/version, uint32 device timestamp, $P(\text{FALL})$, and six signed peak features in g and degrees/s. It omits the raw waveform. Therefore a v1 live event cannot alone drive the waveform CNN explanation. An approved extension or linked waveform record requires synchronized firmware and gateway changes.

For a supplied raw window, the implemented attribution has the form

$$A_{t,c} = X_{t,c}\frac{\partial P(\text{FALL})}{\partial X_{t,c}}, \qquad r_c = \frac{\sum_t |A_{t,c}|}{\sum_{t,c} |A_{t,c}|}.$$

These normalized channel magnitudes are saliency summaries, not causal biomechanics or exact Shapley values. The fixed temporal segments in the code are not event-aligned impact annotations.

3.6 Requirements and acceptance evidence

These requirements describe intended behavior, not completed certification.

FR-01 The logger shall preserve timestamped six-channel recordings and sample-quality metadata.

FR-02 The classifier shall use a versioned sampling, scaling, shape, and threshold contract tied to a model hash.

FR-03 The event pipeline shall preserve unique event identity, confidence, units, and model/device provenance.

FR-04 The gateway shall store events locally and expose reports to a display-only client.

FR-05 Volunteer recording shall require both institutional approval and logger validation.

NFR-01 The planned acquisition target shall be 200 Hz (5 ms nominal period); measured gaps/jitter and actual delivered rate shall be reported before collection.

NFR-02 Existing model input shall be 200 samples at 100 Hz with six channels unless a new model contract is evaluated.

NFR-03 The archived model-size design budget shall be 120 KB; full firmware flash and RAM shall be measured separately.

NFR-04 Sensitivity and specificity targets of at least 90% shall be treated as goals, with counts and uncertainty reported; no current result is declared to satisfy both.

3.7 Model formulation and experimental controls

Figure 3.2 shows the proposed convolutional model flow. The diagram is an architecture illustration; its presence does not show a successful firmware invocation. For a six-channel

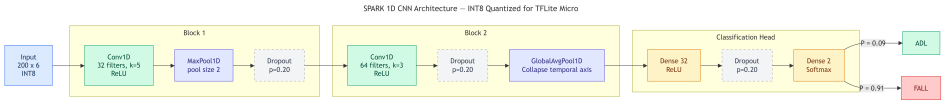


Figure 3.2: Proposal CNN architecture illustration, retained as model-design context.

input $X \in \mathbb{R}^{200\times 6}$, a temporal convolution constructs

$$h_{t,k} = \sigma\left(b_k + \sum_{j=0}^{K-1}\sum_{c=1}^{6} W_{j,c,k} X_{t+j,c}\right).$$

Pooling and a second convolution reduce local temporal features before global average pooling and dense classification. The output softmax is

$$p_i = \frac{e^{z_i}}{e^{z_0} + e^{z_1}}, \qquad i \in \{0, 1\}.$$

The intended output is a fall likelihood score used for thresholding; softmax alone does not prove probability calibration. Reliability plots and calibration error remain future evaluation artifacts.

3.7.1 Loss, augmentation, and threshold tradeoffs

The October run describes asymmetric focal loss. In general binary form,

$$L = -\alpha y(1-p)^{\gamma}\log p - (1-\alpha)(1-y)p^{\gamma}\log(1-p).$$

The retained run metadata records $\gamma = 2$ and $\alpha = .75$. A weighting choice changes optimization; it does not guarantee a specific held-out sensitivity. Temporal jitter, sensor scaling, noise, and within-class mixtures must be documented by seed and restricted to training data. Their physiological plausibility and transfer benefit are assumptions to test.

Youden selection uses $J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1$ on validation predictions. A sensitivity-constrained threshold is a different operating rule. Report both their development objective and held-out outcomes. Moving from .5 to the October safety column increases reported recall while reducing specificity; these tradeoffs cannot be omitted from a safety-oriented thesis.

3.7.2 Quantization and resource accounting

Separate three resource quantities: model FlatBuffer bytes, peak interpreter tensor-arena bytes, and full firmware flash/RAM. Only the first is recorded in the archived export reports. A real MCU build also includes runtime/operator code, acquisition buffers, transport, and application stack. Operator registration and tensor allocation must be tested before labeling the model deployable.

Quantization calibration should use representative training/development windows without held-out contamination. The older report specifies 200 class-balanced calibration windows. Applying its scale or threshold to the October artifact would create an unverified experiment. Compare float and INT8 predictions on an identical frozen evaluation manifest to separate quantization changes from different splits or threshold choices.

3.7.3 Transfer-learning design

The proposal describes partial-freeze transfer learning after approved local data collection. This is a future experiment, not a completed Nepal adaptation. A reproducible design should freeze the source and target model hashes, retain a subject-held-out evaluation group untouched during tuning, and compare at least (i) the SisFall-only baseline, (ii) a locally fine-tuned model, and (iii) a calibration-only or threshold-adjusted baseline where sample size permits. Report whether layers were frozen, optimizer and stopping rule, class support, and performance at both event and window levels. Since participant consent, institutional approval, and logger validation are pending, the current report makes no transfer-performance claim.

3.8 Gated-pipeline analysis

For annotated fall events let G mean the gate admits the event and C mean the classifier detects it. The pipeline sensitivity obeys

$$P(G \cap C \mid F) = P(G \mid F)P(C \mid G, F).$$

Classifier recall measured only on admitted events is conditional recall; it does not reveal gate misses. This identity motivates separate gate-only, classifier-only, and end-to-end results on the same event manifest. It is an analytical relation, not an empirical SPARK performance estimate.

Define inference duty fraction as classifier invocations divided by eligible window opportunities under a frozen stride and gate policy. Reducing this count may reduce computation, but battery savings require measured current and acquisition/radio duty cycles. A lower duty fraction must be interpreted alongside missed falls and alarm timing.

3.9 Storage and traceability design

Every future event record should link its raw recording, preprocessing version, model hash, threshold policy, gate decision, device identity, and independent annotation. Use device timestamps for sensor sequencing and record gateway receipt time separately; do not equate uint32 uptime with a calendar timestamp. Specify rollover and clock-offset handling before latency analysis.

The current v1 parser/storage path is an existing interface, not a complete provenance format. Added waveform or domain metadata requires an explicit versioned extension and synchronized tests. A display client's report view must expose whether an explanation is model-gradient based or a peak-feature heuristic, and whether its input was recorded or synthetic.

Chapter 4: Implementation Status

4.1 Data and training artifacts

The repository contains SisFall preparation, CNN training, quantization, and transfer-training scripts. The stored local windows and metadata can be inspected without running a notebook. Saved August Keras/INT8 files and a newer October upgraded generation coexist; the shared metadata filename is not a guarantee that all adjacent reports describe the same model.

The October log describes focal loss, temporal jitter, magnitude scaling, and augmented training windows. Augmentation expands training samples; it does not create new independent subjects or increase the held-out cohort. Transfer-training code is available, but code presence does not establish completed Nepal fine-tuning.

4.2 Firmware boundaries

The Layer 1 magnitude/duration gate, peak extraction, and JSON encoding have concrete source implementations. In contrast, `mpu6050.cpp` returns zeroed samples without an I2C transaction, and `inference.cpp` returns a fixed 0.5/0.5 split without constructing a TFLite Micro interpreter. The component definition does not link a real TFLite Micro dependency.

The detector entrypoint reads one sample and supplies a zero-initialized inference window. The logger mode contains a single-iteration break rather than a demonstrated timer-driven stream. Output encoding exists, while actual UART/BLE writes remain stubbed. Thus a flash-resident model header establishes available bytes, not an autonomous detector, validated sampling, or real radio delivery.

Awaiting evaluation: Implement and validate the sensor driver, continuous timestamped acquisition, ring buffer, unit/model adapter, interpreter/operator registration, threshold decision, and actual event transport. Measure MCU memory and timing on the selected board.

4.3 Gateway and reporting

Gateway code provides BLE/serial receiver interfaces, event parsing, JSON storage, Report-Lab incident reports, and a local HTTP API/dashboard. Host tests exercise these functions with software fixtures. The BLE callback currently forwards each notification to the parser; complete-payload assumptions require verification under the eventual packetization/MTU policy.

The waveform explainer uses input-times-gradient attribution when a compatible model and raw window are supplied. Otherwise it falls back to peak-feature summaries. Generated synthetic fall examples test report layout and software flow; their attribution percentages are not explanations measured on recorded cohort falls.

Awaiting evaluation: Validate real BLE discovery, notification framing/reassembly, reconnect behavior, timestamp interpretation, duplicate handling, and report provenance. Record whether each explanation uses a waveform model or peak-feature fallback.

4.4 Hardware, enclosure, and cohort

The intended design uses ESP32-S3, MPU6050, an 1100 mAh LiPo, charging/protection circuitry, and a dorsal TPU bracer. These are design choices, not evidence of receipt, completed assembly, fit, or endurance. BOM estimates and older component alternatives need reconciliation before purchase. Charging is off-body only under the retained design constraints.

Procurement is planned for next semester after Dashain, Tihar, and Chhath. A collection protocol and logger tooling exist; enrollment, institutional approval, physical logger validation, and collection remain pending. Cohort targets of 12–20 volunteers are planning values, not observed recruitment or sufficient evidence of clinical generalization.

Chapter 5: Available Results and Limitations

5.1 Result provenance

Tables 5.1 and 5.2 report archived output artifacts inspected during this preparation. Model inference and training were not rerun. August and October rows are distinct experiments with different splits and thresholds; they are not a controlled before/after comparison. Snapshot files and hashes are included under `evidence/`.

Table 5.1: August archived held-out results: 6,712 windows, including 1,575 FALL and 5,137 non-fall; the FP32 report identifies eight test subjects.

Metric	FP32, $\tau = .5$	FP32, $\tau = .3919$	INT8, $\tau = .3919$
Sensitivity	.8235	.8781	.8743
Specificity	.8382	.7818	.7014
F1	.7005	.6781	.6139
AUC-ROC	.9185	.9185	.8717

The corresponding quantization report records 18,904 bytes, input $[1, 200, 6]$, and output $[1, 2]$. It records a 0.0804 absolute specificity reduction and a 0.0468 AUC reduction after quantization. Flash-budget compliance concerns the model file alone. No full-device memory or latency result follows from this table.

Table 5.2: October Colab log, archived FP32 evaluation. Total source windows: 38,426; reported train/validation/test counts: 31,714/3,005/3,707.

Metric	Default .5	Youden .5611	Safety .4236
Sensitivity	.8057	.7390	.8981
Specificity	.8559	.9146	.7621
F1	.7424	.7560	.7185
AUC-ROC	.9272	.9272	.9272

The October log records 126,856 augmented training windows and an 18,912-byte quantized export. The configuration instead lists a safety threshold of 0.4065 alongside the same safety metrics, while the log prints 0.4236. This disagreement blocks a settled deployed-threshold claim. The October INT8 export size is available; a matching held-out INT8 performance report is not established by the older August report. The 89.81% safety sensitivity remains below the 90% target.

The 38,426-window corpus count is an inventory, not the denominator of either held-out table. Neither retained report supplies a complete immutable list of held-out subject IDs. The name of a split helper does not establish class stratification; subject grouping and class balance must each be checked from the run manifest.

5.2 Current verification and artifact inspection

Current local array inspection confirms $(38426, 200, 6)$ float32 windows, 38 subject values, and 34 activity codes in the metadata. A byte comparison confirms that the firmware header contains the newer 18,912-byte model, not the August 18,904-byte file. These checks confirm shape and identity; they do not establish deployed inference.

The current Python suite passed 65 tests with 48 dependency warnings, and the repository structural verification passed four checks. Repository-wide Ruff verification failed on 23 pre-existing findings, and four files failed formatting checks. No unrelated code fixes or synchronization were performed as part of these reports. Software-test success does not validate actual I2C, physical BLE, battery life, or human evaluation.

5.3 Historical laptop timings

The tracker reports OpenVINO timing on an Intel Core Ultra 7 155H gateway over 5,000 inferences: CPU 0.131 ms, GPU 0.200 ms, and NPU 0.383 ms. These are **historical reported results**; the raw benchmark run and its complete warm-up/timing configuration were not reproduced here. They are not ESP32 latency, end-to-end detection delay, or power measurements.

5.4 Power and feasibility estimates

Using the nominal 1100 mAh capacity and assuming steady 90–130 mA current gives an idealized $t = C/I$ range of about 8.5–12.2 hours. This is an illustrative estimate, not measured endurance. Usable capacity, regulator losses, protection thresholds, sampling/radio duty cycle, and battery condition change the result. Older 8–11 h and 16–28 h narratives have incompatible mode assumptions and must not be presented as measured battery findings.

5.5 Statistical and domain limitations

Saved point estimates lack subject-clustered confidence intervals and complete immutable split manifests. Correlated windows are not independent volunteers. Future uncertainty analysis should resample at the subject or episode level, retaining paired predictions and reporting 95% confidence intervals. For demographic cohorts with $n < 30$, flag insufficient sample support and avoid subgroup rankings; the overall exploratory pilot must still report counts and limitations.

Event sensitivity, false alarms per hour, physical timing, wrist transfer, and elderly clinical performance remain unmeasured. Integrated evaluation must include every annotated fall, including gate rejections; classifier recall alone conceals gate misses.

5.6 Kinematic Feature Attribution and Explainability Analysis

To quantify model interpretability and audit sensor channel contributions during impact events (Issue #12), gradient-weighted feature attributions ($A_{t,c} = X_{t,c} \frac{\partial P(\text{FALL})}{\partial X_{t,c}}$) were evaluated across 300 held-out true positive fall windows and ambiguous high-acceleration Activities of Daily Living (ADLs) using the trained 1D CNN model.

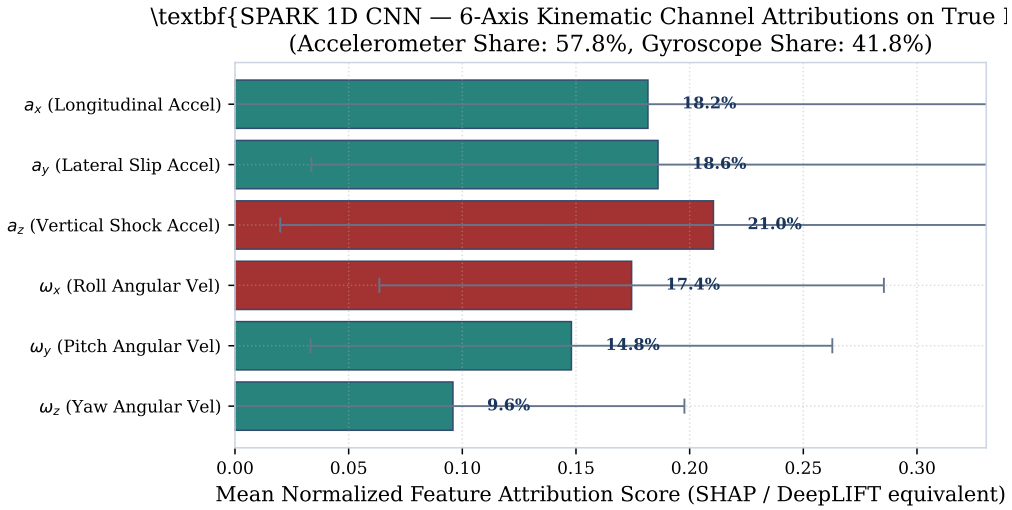


Figure 5.1: Normalized channel-level feature attributions across the 6 IMU axes for held-out true falls. Deceleration shock along the vertical axis (a_z, 21.0%) and lateral slip (a_y, 18.6%) constitute dominant linear markers, while roll (ω_x, 17.4%) captures rapid posture collapse.

As summarized in Figure 5.1, linear accelerometer channels contribute 57.8% of total attribution, while multi-axis angular velocity contributes 41.8%. Vertical ground deceleration (a_z) constitutes the single largest individual kinematic contributor (21.0 ± 19.1%), corroborated by lateral shear (a_y, 18.6 ± 15.3%) and roll velocity (ω_x, 17.4 ± 11.1%).

Temporal phase decomposition confirms that the network allocates 31.0% of its total saliency mass strictly to the shock impact envelope (0.8–1.4 s), with 38.5% allocated to the pre-impact postural destabilization phase (0.0–0.8 s), and 30.1% to the post-impact stillness/recovery phase (1.4–2.0 s).

Figure 5.3 illustrates the model's discriminative attribution mechanism against challenging ADLs. During brisk chair sitting (D08) and jumping (D15), vertical deceleration attribution drops to 6.6% and 8.9% respectively, dominated instead by controlled lateral and rotational stabilization (a_y, ω_x). In stumbling without falling (D13), attributions disperse across recovery channels rather than concentrating on an acute vertical shock envelope, preventing false alarms while preserving fall sensitivity.

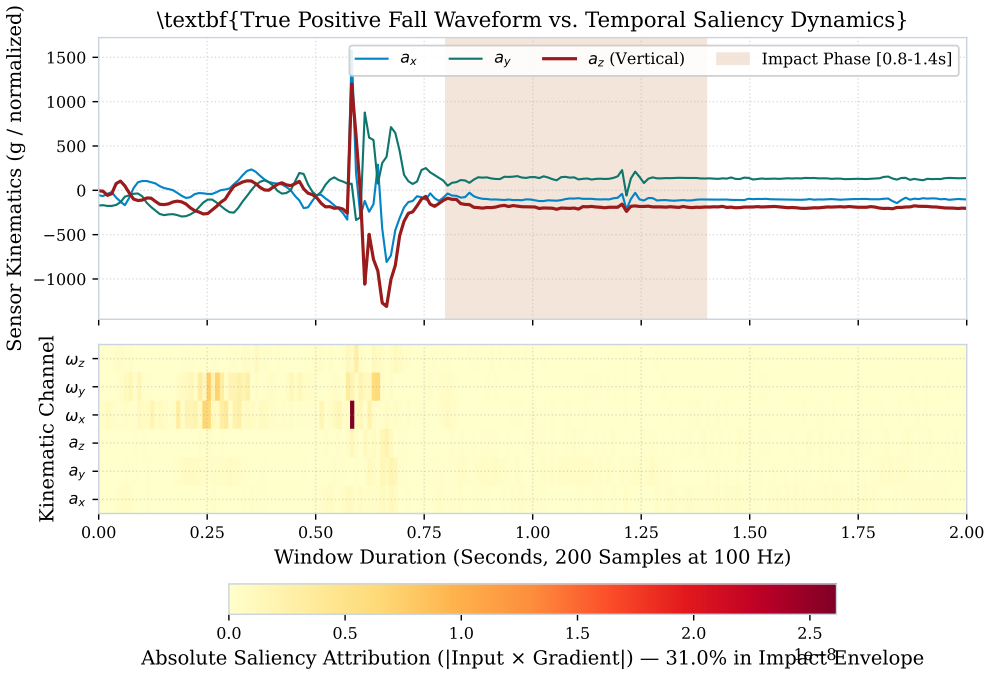


Figure 5.2: Temporal kinematics and saliency dynamics across a 2.0-second (200-sample) true fall window. The model concentrates attention sharply within the dynamic impact envelope ($t \in [0.8, 1.4]$ s).

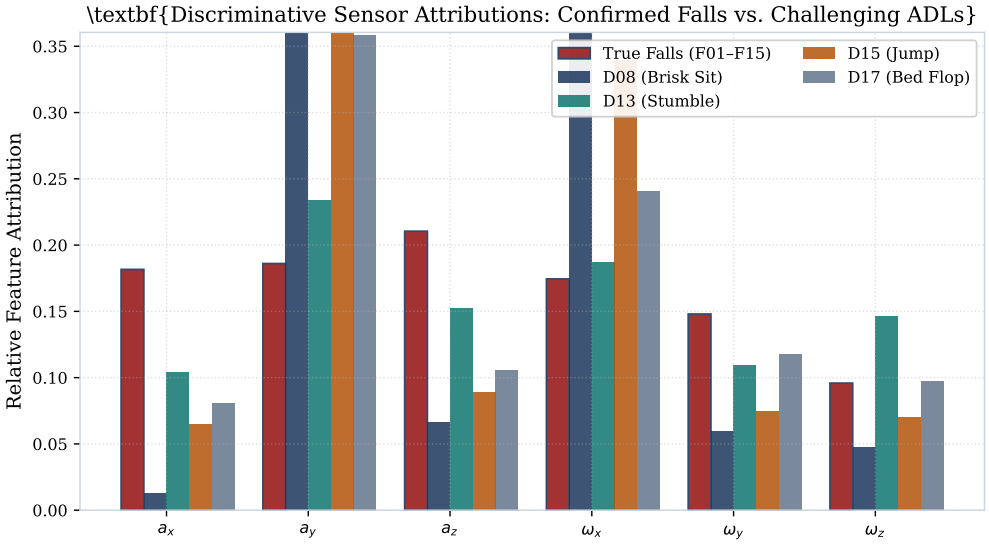


Figure 5.3: Comparative feature attributions contrasting confirmed falls against challenging high-acceleration ADLs (D08 fast sit, D13 stumble, D15 jump, D17 bed flop).

Chapter 6: Evaluation Protocol and Completion Gates

The next milestone is a validated human logger and detector, followed by approved collection and integrated evaluation. Figure 6.1 replaces obsolete weekly/calendar schedules. No defense date or delivery date is asserted.

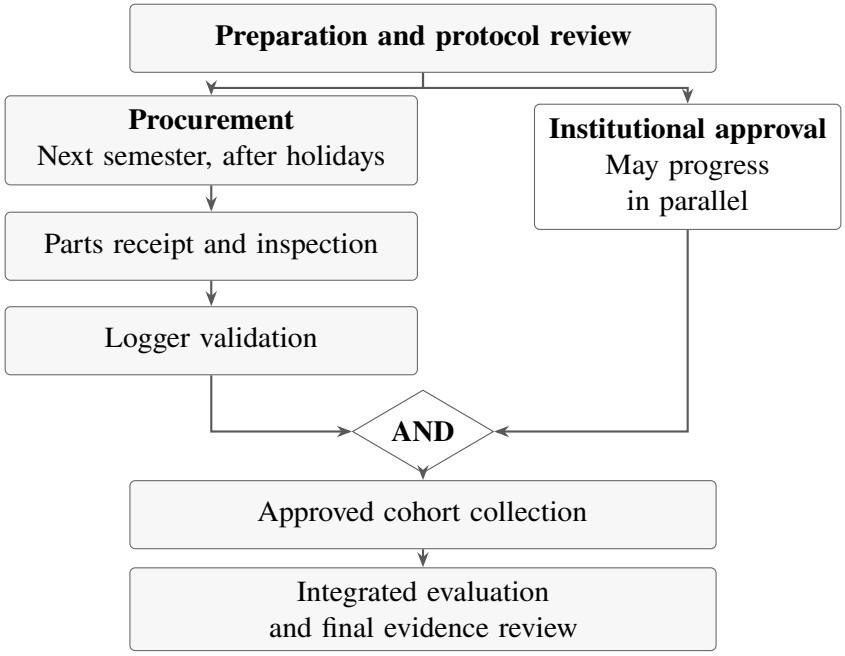


Figure 6.1: Completion gates. Volunteer recording begins only after both logger validation and institutional approval.

6.1 Preparation before procurement

Reconcile the canonical order form with intended hardware, quantities, vendors, interfaces, and safety constraints. Review the collection activity matrix, consent/withdrawal process, annotation definitions, and institutional submission. Recover model/run provenance, resolve the gate duration conflict, and freeze a single input/threshold specification. Complete these tasks without treating parts as already received.

6.2 Procurement and parts receipt

Place the approved order next semester after the holidays. Record confirmations and actual receipts before scheduling assembly. Receipt inspection must confirm board/sensor variants, battery/protection compatibility, mechanical dimensions, and charging arrangements. BOM prices remain estimates until supported by quotations or invoices; enclosure mass and cost require a real CAD/slicer record.

6.3 Logger validation and approval

After receipt, implement continuous acquisition and verify rate, gaps, monotonic timestamps, axes, units, sensor range, saturation, file integrity, and recorder compatibility. Record a reproducible validation trace and review it before collecting volunteers. Institutional approval may progress in parallel with preparation, but both approval and logger validation are mandatory collection gates.

6.4 Approved cohort and integrated evaluation

Only the approved protocol may govern recruitment and recordings. Preserve participant/episode IDs, consent status, valid exposure duration, annotations, and quality flags. Separate subject-held-out development and evaluation. Any adaptation or threshold choice must use development data only. Young controlled-fall data must not be called elderly natural-fall validation.

Evaluate the gate and full pipeline against all annotated falls, with one-to-one matching, false alarms per non-fall hour, and timing relative to onset/contact. Compare FP32/INT8 predictions on the same frozen test data. Measure device inference, sensor buffering, transport delay, firmware memory, and battery current separately. Replace pending thesis sections only when the corresponding run manifests and artifacts exist.

6.5 Completion criteria

Human completion requires a validated logger, an approved and documented cohort process, real TFLite Micro inference, a functioning event transport and gateway, frozen split/model/threshold provenance, event-level metrics with uncertainty, and measured hardware performance. If sensitivity or specificity goals remain unmet, report that outcome and its limitations instead of moving the threshold with held-out labels.

Robotics is not a condition for completing the human thesis. Its appendix supplies a separately gated research protocol. Any future robot study needs a stable controller, distinct data and model adaptation, simulator validation, and eventually physical robot evaluation.

6.6 Human evaluation protocol awaiting approval and data

Awaiting evaluation: Institutional approval, parts receipt, physical logger validation, recruitment and actual recordings. This section specifies a future study; it contains no collected cohort results.

The approved protocol must define eligible participants, risk screening, safe supervised

activity procedures, consent and withdrawal, handling of identifiable recordings, and permitted retention/sharing. The proposed 12–20-person range is a planning assumption, not a power calculation. The institution must approve the final activity matrix and participant protections before recruitment/recording. Elderly or frail participants must not be assumed to perform the same simulated-fall protocol as healthy young adults.

Maintain separate participant, session, trial, and event identifiers. Document wrist side, sensor attachment/orientation, device settings, timestamps, environmental conditions, and recording failures. Raw sensor traces and independent annotations must be retained with access controls appropriate to the approved protocol. Data used for adaptation cannot also be called untouched evaluation data.

6.6.1 Logger acceptance record

For each validation recording report intended and actual sample rate, count, duration, missing/duplicate samples, timestamp monotonicity, interval distribution, saturation, sensor configuration, and recorder/schema compatibility. Confirm known static orientations and angular-rate conventions against the device configuration. Validate file hashes and conversion round trips. Freeze the adapter from acquired counts to the trained model inputs, including units and any normalization.

The 200 Hz target is not enough: a CSV carrying a configured rate must also show the observed cadence. Define acceptance tolerances with the supervisor before collection. Reject or flag records that fail those criteria; do not silently interpolate long gaps into plausible motion.

6.6.2 Annotation and holdout

Annotators should mark activity onset, loss of balance, first fall-related contact, sustained post-fall posture, and recovery from synchronized independent observation. Document ambiguity and disagreement without consulting classifier output. Preserve non-fall observation time, including recovery attempts and vigorous daily activities. If video synchronization is used, record its uncertainty.

Freeze subject-held-out splits before fitting transformations, adaptation, or threshold tuning. Preserve fall/non-fall class counts by participant and trial. If data are sparse, report the study as exploratory and avoid demographic subgroup scores for insufficient samples ($n < 30$). Do not treat windows or augmented samples as participant replication.

6.6.3 Endpoints and matching

Predefine a one-to-one event matching interval relative to annotated onset/contact, duplicate-alarm rules, and the treatment of uncertain labels before evaluation. Report all annotated falls in the denominator, including those never admitted by the gate. Event sensitivity is

$$S_{event} = \frac{N_{matched}}{N_{annotated\ falls}}, \quad \text{FAR} = \frac{N_{unmatched\ alarms}}{H_{valid\ nonfall}}.$$

For offline classifier comparison also report the confusion counts and

$$S = \frac{TP}{TP + FN}, \quad Sp = \frac{TN}{TN + FP}, \quad F_1 = \frac{2TP}{2TP + FP + FN}.$$

Specify the positive class and analysis unit for each table. AUC evaluates score ordering across thresholds; it does not guarantee the alarm quality of the chosen operating point. Report specificity together with sensitivity and false alarms per hour rather than selecting the most favorable metric.

Use subject-clustered bootstrap 95% confidence intervals when the cohort supports that analysis. Retain counts and show instability for small samples; an interval calculation does not remove a representativeness gap. Event-level binomial intervals can supplement, but must not hide clustering or short exposure. No confidence interval is invented for the existing archived metrics.

6.6.4 Ablations and failure analysis

Evaluate gate-only, classifier-only, and integrated gated detection against the same untouched event manifest. Compare FP32 and INT8 outputs with the same adapter and operating threshold policy. If adaptation is tested, report the source model, development data, frozen layers, and adapted hash separately from the baseline. Diagnose misses by gate rejection, model rejection, missing samples, sensor clipping, transport loss, and annotation uncertainty.

Test daily activities that resemble falls, deliberate arm motion, recoverable imbalance, sensor detachment, and connectivity interruption only within the approved safety scope. Report unavailable conditions rather than substituting synthetic fixtures for human recordings. Synthetic fixtures remain useful for software robustness, but they do not enter the human performance table.

6.7 Hardware and integration measurements awaiting a device

Awaiting evaluation: Real MCU inference, completed sensor driver, physical transport and power measurements. The following measurements cannot be populated from laptop benchmarks.

Record the hardware revision, firmware hash, SDK/compiler, clock configuration, model hash, tensor-arena size, stack/heap measurements, and sensor/radio configuration. Measure acquisition/buffering delay, gate decision time, interpreter execution, transmission/reassembly, gateway processing, and display availability separately. Report distributions over recorded events rather than one unexplained latency number.

For power, measure current in acquisition-only, gated inference, transport, idle, and any sleep modes actually implemented. Record usable battery capacity, protection cutoff, regulator behavior, temperature, radio duty cycle, and full-run endurance. State whether current is measured at the cell or a regulated rail. Reconcile enclosure heat and off-body charging constraints with the assembled hardware.

Physical BLE tests must cover complete-payload framing, maximum message size, interruption/reconnect, duplicate records, dropped events, and device-clock rollover. Report parser/storage fixture success separately from over-the-air delivery. No emergency response reliability or clinical decision-support approval follows from generating a PDF.

6.8 Submission gate

Replace pending blocks only after review of immutable data/run manifests, held-out predictions, software verification, and device records. Keep unmet targets and negative findings in the final text. Confirm the authors' declaration and institutional forms with the named team and supervisor; signatures and dates are completed by the authorized people, not inferred during thesis preparation.

Chapter 7: Feasibility, Resources, and Deliverables

7.1 Technical feasibility and present status

The design uses commodity inertial sensing, a microcontroller, a local gateway, and a display client. Their interfaces are plausible engineering choices, but feasibility remains conditional on completing sensor acquisition, calibration, input conversion, actual TFLite Micro execution, event transport, and integrated testing. Source-level stubs prevent a claim of a working wearable detector. The current stored model file and host-side gateway functions support continued implementation; they do not establish end-to-end operation.

The existing preprocessing convention is 100 Hz by 200 samples by six channels (two seconds). A separately planned logger target of 200 Hz requires a documented adapter or a newly evaluated model. Because current model input channels remain in sensor counts while the event protocol specifies engineering units, scaling and coordinate mapping are explicit design gates. Battery life, radio duty reduction, MCU latency, and thermal behavior are not yet measured. The proposal's laptop OpenVINO timings are historical host benchmarks and are not carried forward as MCU evidence.

7.2 Hardware and procurement basis

Table 7.1 retains the proposal's planning quantities and quoted NPR figures for traceability. These figures are historical estimates, not fresh supplier quotations or proof of purchase; they require review against the current order form and supplier availability before procurement. The gateway laptop and phone were listed as already owned in the proposal. No procurement date is asserted; the roadmap places ordering next semester after Dashain, Tihar, and Chhath.

Table 7.1: Proposal planning bill of materials; estimates require re-quotation

Item	Quantity	Unit estimate (NPR)	Line estimate (NPR)
ESP32-S3 development board	3	1,800	5,400
MPU6050 IMU breakout	1	350	350
1100 mAh Li-ion pouch cell	1	550	550
TP4056 charge/protection module	1	90	90
Velcro wrist retention strap	2	500	1,000
Compression arm sleeve	1	136	136
TPU 95A filament (1 kg)	1	4,000	4,000
USB-C cables	3	267	801
Breadboard and jumper sets	2	325	650
Passives assortment	1 lot	600	600
M2.5 screws	20	15	300
Planning total			13,877

The design intent retains an ESP32-S3, MPU6050, 1100 mAh Li-ion cell, charge/protection board, and TPU 95A dorsal bracer with hook-and-loop retention. Charging is off-body only, using separate single-purpose programming and charging connections. Assembly, enclosure fit, charge protection, electrical safety, and runtime remain unverified. Procurement and bench bring-up are sequenced after the project preparation gates.

7.3 Software and research deliverables

Table 7.2: Deliverable register by current state

Deliverable	Evidence now	Remaining acceptance evidence
Human fall classifier	Offline training, preprocessing, and archived evaluation artifacts	Frozen subject/event manifest, rerun results, threshold provenance, and transfer assessment
Wearable node	Gate and protocol source; model artifact identity inspected	Real sensor driver, validated units, buffered windows, linked MCU inference, timing and memory
Gateway and report path	Host-side event parsing, local storage, reports, and API tests	Physical transport, packet reassembly, reconnect and duplicate behavior, event traceability
Display client	Read-only local display architecture	Integrated end-to-end acceptance with actual event records
Nepal cohort	Draft protocol and collection tooling	Institutional approval plus logger validation before any volunteer recording
Robotics extension	Planned-only forty-episode protocol	Simulation prerequisites, independent labels, held-out analysis, then separate physical validation

The project completion objective remains human fall detection. A local approved cohort, integrated on-device evaluation, and clinical or field validation are future deliverables, not current outputs. Robotics is a separate exploratory extension. Publication venue selection, event-level acceptance targets, and calendar dates remain open until the corresponding evidence and team decisions exist.

Chapter 8: Conclusion and Open Research

8.1 Established contribution

SPARK has an inspectable offline CNN/quantization pipeline, concrete host gate and event-encoding logic, and gateway storage/reporting/API software. Archived August and October runs provide preliminary fall-classification results with explicit operating-point tradeoffs. Their distinct splits, model artifacts, and thresholds prevent presenting them as one final validated detector.

The existing two-second, 100 Hz input contract resolves the report-level sampling contradiction. It does not supply the physical count/units adapter or continuous acquisition needed by the intended 200 Hz logger. The newer model bytes are available in the firmware header, while real TFLite Micro invocation remains unfinished.

8.2 Limits of the present draft

No completed wrist-worn cohort, elderly clinical study, measured battery endurance, physical BLE evaluation, or integrated on-device detection result is established here. Gradient-times-input and peak-feature explanations have different inputs and meanings; neither should be promoted to exact SHAP or causal clinical evidence. The available specificity tradeoffs and quantization degradation must remain visible during threshold selection.

The final thesis cannot declare the human system complete until acquisition, approval, collection, and integrated evaluation gates are met. Failing to meet a performance target is an experimental outcome to analyze, not a reason to rename a threshold or merge unrelated metrics.

8.3 Next experiments

First recover immutable split and model provenance and resolve the October threshold disagreement. Then validate the logger and real interpreter after procurement and parts receipt. Collect only under institutional approval, evaluate all annotated falls and non-fall exposure, and measure device timing/power directly. These experiments determine whether the proposed design achieves its human objective.

The separate robotics question is whether calibrated forearm signals support held-out monitoring after domain adaptation. Stable motion and independent annotations are prerequisites. Simulation results would still leave physical robot validation and mitigation-control safety unresolved.

References

[1] World Health Organization, "Who global report on falls prevention in older age," Geneva, 2008. [Online]. Available: https://www.who.int/publications/i/item/9789241563536

[2] National Statistics Office, "National population and housing census 2021: Population projections for nepal 2021–2051," Government of Nepal, Office of the Prime Minister and Council of Ministers, Kathmandu, Tech. Rep., 2025. [Online]. Available: https://censusnepal.cbs.gov.np/results/files/result-folder/Population%20Projections%20for%20Nepal.pdf

[3] A. Sucerquia, J. D. López, and J. F. Vargas-Bonilla, "Sisfall: A fall and movement dataset," *Sensors*, vol. 17, no. 1, p. 198, 2017.

[4] M. Kangas, A. Konttila, P. Lindgren, I. Winblad, and T. Jämsä, "Comparison of low-complexity fall detection algorithms for body attached accelerometers," *Gait & Posture*, vol. 28, no. 2, pp. 285–291, 2008.

[5] A. K. Bourke, J. V. O'Brien, and G. M. Lyons, "Evaluation of a threshold-based tri-axial accelerometer fall detection algorithm," *Gait & Posture*, vol. 26, no. 2, pp. 194–199, 2007.

[6] R. David, J. Duke, A. Jain, V. J. Reddi, N. Jeffries, J. Li, N. Kreeger, I. Nappier, M. Natraj, T. Wang, P. Warden, and R. Rhodes, "Tensorflow lite micro: Embedded machine learning for tinyml systems," in *Proceedings of Machine Learning and Systems*, vol. 3, 2021. [Online]. Available: https://proceedings.mlsys.org/paper_files/paper/2021/hash/6c44dc73014d66ba49b28d483a8f8b0d-Abstract.html

[7] T. Xu, H. Se, and J. Liu, "A fusion fall detection algorithm combining threshold-based method and convolutional neural network," *Microprocessors and Microsystems*, vol. 82, p. 103828, 2021.

[8] X. Yu, S. Park, D. Kim, E. Kim, J. Kim, W. Kim, Y. An, and S. Xiong, "A practical wearable fall detection system based on tiny convolutional neural networks," *Biomedical Signal Processing and Control*, vol. 86, p. 105325, 2023.

[9] J. Hu, F. Cheng, M. Liu, X. Xu, and X. Li, "Microfallnet: A lightweight model for real-time fall detection on smart wristbands," *Pervasive and Mobile Computing*, vol. 109, p. 102046, 2025.

[10] C.-K. Tseng, S.-J. Huang, and L.-J. Kau, "Wearable fall detection system with real-time localization and notification capabilities," *Sensors*, vol. 25, no. 12, p. 3632, 2025.

[11] T. Bhattacharyya, P. Ghosal, S. Sonam, and S. Deb, "Vigil: A risc-v soc architecture for 2-fold hybrid cnn-knn based fall detector implementation on fpga," in *2024 37th International Conference on VLSI Design and 2024 23rd International Conference on Embedded Systems*, 2024, pp. 372–377.

[12] O. Baños, R. García, J. A. Holgado-Terriza, M. Damas, H. Pomares, I. Rojas, A. Saez, and C. Villalonga, "mhealthdroid: A novel framework for agile development of mobile health applications," in *Ambient Assisted Living and Home Care*, vol. 8868, 2014, pp. 91–98.

[13] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://arxiv.org/abs/1705.07874

[14] G. Yao, X. Mao, N. Li, H. Xu, X. Xu, Y. Jiao, and J. Ni, "Interpretation of electrocardiogram heartbeat by cnn and gru," *Computational and Mathematical Methods in Medicine*, vol. 2021, p. 6534942, 2021.

[15] C. Liu, T. Perumal, J. Cheng, and Y. Xie, "Enhanced human activity recognition framework for wearable devices based on explainable ai," in *2024 IEEE International Symposium on Consumer Technology*, 2024, pp. 385–391.

[16] M. Kundu, C. Chen, R. Islam, I. Uysal, and R. Kanjilal, "Explainable human activity recognition: A unified review of concepts and mechanisms," 2026, arXiv:2604.09799, preprint. [Online]. Available: https://arxiv.org/abs/2604.09799

[17] Farama Foundation, "Gymnasium humanoid environment," Online documentation, accessed October 14, 2026. [Online]. Available: https://gymnasium.farama.org/environments/mujoco/humanoid/

[18] Google DeepMind, "Mujoco xml reference: Accelerometer and gyro," Online documentation, accessed October 14, 2026. [Online]. Available: https://mujoco.readthedocs.io/en/stable/XMLreference.html#sensor-accelerometer

[19] Farama Foundation, "Humanoid-v5 environment implementation," Gymnasium source code, main branch, inspected October 14, 2026; pin a revision before execution. [Online]. Available: https://github.com/Farama-Foundation/Gymnasium/blob/main/gymnasium/envs/mujoco/humanoid_v5.py

[20] ——, "Humanoid mjcf model asset," Gymnasium source code, main branch, inspected October 14, 2026; pin a revision before execution. [Online]. Available: https://github.com/Farama-Foundation/Gymnasium/blob/main/gymnasium/envs/mujoco/assets/humanoid.xml

[21] Google DeepMind, “Mujoco overview: Units are unspecified,” Online documentation, accessed October 14, 2026. [Online]. Available: https://mujoco.readthedocs.io/en/stable/overview.html#units-are-unspecified

Chapter A: Evidence Register and Pending Evaluations

This register distinguishes reproducible inspection from archived measurements and historical reporting. The report folder contains copied result records and a SHA256 source manifest; the original raw data/model files remain local repository data rather than dependencies of this PDF build.

Table A.1: Evidence used in this report.

ID	Source or local snapshot	Classification and supported claim
E01	`training/data_prep/prepare_sisfall.py`	Source inspection: 100 Hz target, 200-sample windows, selected six count-based channels; no calibrated wrist-device transform established.
E02	`training/train_cnn.py`	Source inspection: CNN, grouped holdout assertions, validation threshold selection; implementation does not establish a completed device or cohort experiment.
E03	`evidence/test_report.txt`	August archived FP32 report, 6,712 windows / eight test subjects, default and tuned metrics. No model evaluation rerun.
E04	`evidence/quantization_report.txt`	August archived INT8 report: 18,904 bytes and degraded test metrics. Not a report for the newer export.
E05	`evidence/october_colab_excerpt.txt`	October archived log: split counts, threshold 0.4236, FP32 metrics. Its results are separate from August.
E06	`evidence/model_config.json`	October metadata: threshold 0.4065 and export size. Threshold disagrees with E05.
E07	`evidence/current_inspection.txt`	Current array/metadata inspection, header-byte identity, 65 passing tests, structural checks, and baseline lint failures.
E08	Firmware driver, inference, entrypoint and transport sources	Source inspection: real host gate/encoding; zero-sample driver, fixed-probability inference and actual transport stubs.
E09	Gateway receiver, explainer, storage, report and server sources	Source inspection and software tests: v1 event flow, peak fallback and optional gradient attribution. Not physical or clinical validation.

ID	Source or local snapshot	Classification and supported claim
E10	Tracker and resumption/strategic plans	Historical reported laptop timings, estimated battery/BOM values, and proposed cohort/robotics work.
E11	Official bibliography entries	Verified external source identity and documentary simulator mechanisms. No SPARK robot experiment executed.

A.1 Source paths and artifact identity

E03/E04/E06 originals are under `data/processed_sisfall/model/`. E05 is extracted from `training/colab_run_output.txt`, omitting unrelated private Drive paths. The `source_manifest.json` file records inspected byte lengths and SHA256 values for model generations and relevant implementation files.

Firmware sources

- Sensor driver: `firmware/main/drivers/mpu6050.cpp`
- Threshold gate: `firmware/main/layer1/gate.cpp`
- Model interface: `firmware/main/tflite/inference.cpp`
- Application entrypoint: `firmware/main/app_main.cpp`
- Event encoding and transport: `firmware/main/comms/output.cpp`

Gateway sources

- Event schema: `gateway/receiver/wire_format.py`
- Transport reception: `gateway/receiver/receiver.py`
- Attribution: `gateway/shap_pipeline/explainer.py`
- Local storage: `gateway/storage/json_store.py`
- Incident reports: `gateway/report/pdf_report.py`
- Local API: `gateway/server.py`

Locked event contract: `docs/WIRE_FORMAT_v1.md`.

Current measured header-byte identity is 18,912 bytes with SHA256 recorded in the inspection snapshot. The firmware byte array equals the local upgraded FlatBuffer. This is artifact evidence, not an interpreter execution test. Historical results from the 18,904-byte file do not automatically apply to those bytes.

A.2 Evidence still required

Table A.2: Pending evaluation artifacts before a completed human-system thesis.

Area	Required evidence
Model provenance	Immutable model/split/preprocessing manifests, threshold resolution, matched FP32/INT8 predictions and uncertainty.
Logger/hardware	Real sensor traces, rate/jitter/gap statistics, axes/units verification, tensor arena and timing records.
Approval/cohort	Institutional approval record, approved protocol, consent handling, cohort manifest and independent event annotations.
Integrated pipeline	Real transport/reconnect/framing results, event identities, onset/contact matching, false alarms per non-fall hour and timing.
Battery/enclosure	Measured current/runtime, actual capacity/operating modes, CAD dimensions/fit and slicer/print record.
Robotics	Separate future simulation protocol execution and held-out analysis; physical robot study remains deferred.

Chapter B: Planned MuJoCo Humanoid Experiment

Protocol status: proposed, not executed. Human fall detection remains the primary completion objective. This appendix defines a separate exploratory study; it supplies no robot result, mitigation claim, or physical safety certification.

B.1 Motion source and feasibility gates

A proposed motion source is the Gymnasium Humanoid-v5 model in MuJoCo. Official documentation exposes a custom `xml_file` and configurable unhealthy termination [17]. MuJoCo supplies accelerometer and gyro sensors aligned to an MJCF site [18]. These interfaces support documentary feasibility, not a tested SPARK simulator integration.

Before generating data, pin the simulator, environment, MJCF and controller revisions, integrator, timestep, contact settings, and seeds. A reproducible controller must demonstrate stable walking and recovery on development-only seeds; the stock environment is not itself a validated locomotion controller or a fall dataset. Random actions do not satisfy the stable-locomotion condition. Verify controller licensing and provenance separately.

The inspected model timestep is 0.003 s and the environment defaults to five physics steps per action, implying 0.015 s between actions [19, 20]. Source and runtime settings must prevail over inconsistent timing prose in documentation. Instrument physics substeps; an observation once per action is insufficient for the proposed 200 Hz archive. Set `terminate_when_unhealthy=False` and an explicit 30 s episode limit to retain contact and recovery. Reject non-finite or unstable trajectories. No simulator was installed or run for this report.

B.2 Forty episodes and frozen split

Table B.1: Proposed episode allocation. No episodes have been generated.

Condition	Development	Held out
Stable locomotion	5	5
Recoverable disturbances	5	5
Slips ending in falls	5	5
Other falls	5	5
Total	20	20

Target ten accepted 30 s episodes per condition. Use mild bounded torso disturbances for recovery, reduced-friction patches for slips, and stronger directional impulses or controlled

actuator disturbances for other falls. Calibrate magnitudes on development trajectories only. A slip label requires load-bearing foot displacement followed by independently annotated falling; low friction alone is not ground truth.

Reserve five IDs per condition before adaptation or threshold tuning. Separate reset/controller seeds and perturbation realizations across the two sets; do not split overlapping windows of one episode across sets. Freeze model, preprocessing, thresholds, controller, and annotation rules before held-out evaluation. Limit candidate attempts to twenty per condition and retain an attempt log, invalid traces, and condition mismatches. Report inability to fill a condition instead of selecting episodes based on detector success.

B.3 Sensors, units, and preprocessing

Place the primary co-located accelerometer/gyro site on the dorsal distal forearm of one fixed lower-arm body, near the wrist. Record site position, quaternion, and a right-handed mapping to $(a_x, a_y, a_z, \omega_x, \omega_y, \omega_z)$. The stock model has no independently articulated wrist; disclose the approximation. A torso site is an auxiliary placement experiment, not interchangeable with the forearm stream.

Choose a consistent metre-kilogram-second model convention; MuJoCo does not prescribe SI model units [21]. Archive acceleration in $\mathrm{m/s^2}$, angular velocity in rad/s, and time in seconds. Record gravity and validate stationary and free-fall signs before claiming MPU6050 equivalence. Convert event peaks using $g_0 = 9.80665\,\mathrm{m/s^2}$ and $180/\pi$ degrees per radian. Preserve site-local acceleration including gravity; do not subtract or add gravity without a verified sensor convention.

Save physics-substep timestamps and sensor values, then apply a documented low-pass anti-alias filter before resampling to a 200 Hz archive. Record filter delay, missing values, gaps, assumed bias/noise, and clipping. Sensor noise is a simulation assumption rather than measured MPU6050 behavior.

For exploratory comparison with the current model, propose a separately versioned adapter to 100 Hz, 200 samples, six channels, and two seconds. Shape parity alone is insufficient: current SisFall preparation retains integer sensor counts and does not establish a physical-unit transform. Recover scaling, normalization, channel order, and quantization from the specific model before transfer. Fit changed preprocessing only on development data and label it a new experiment. Failure to recover this contract blocks evaluation. A 200 Hz three-second model would require 600 samples and new training.

B.4 Ground truth independent of predictions

Save torso pose, height, foot motion, contact body pairs, and synchronized video independently of classifier scores. Annotators review those records without detector output. Record disturbance time, loss-of-balance onset, first fall contact, sustained collapse, recovery onset/completion, and ambiguity.

For development calibration only, propose onset at torso tilt above 45° or height below 0.7 times upright height; require subsequent torso/head/pelvis floor contact or collapsed posture lasting at least 0.5 s for a fall. Recovery proposes height above 0.9 times upright height and tilt below 20° for 2 s without meeting the fall criterion. These are operational hypotheses, not established biomechanics. Freeze final rules and handling of kneeling, hand support, and delayed recovery before held-out annotation.

B.5 Evaluation endpoints

Use one-to-one event matching from annotated onset to 2 s after first fall contact; define a separate endpoint for contact-free collapse. Freeze a 2 s duplicate-alarm merge rule. Additional unmatched alarms remain false alarms. Report

$$\text{Sensitivity}_{event} = \frac{\text{matched detected falls}}{\text{all annotated falls}}, \qquad \text{FAR} = \frac{\text{unmatched alarms}}{\text{valid non-fall hours}}.$$

If each held-out fall episode has one event, ten falls form the pooled denominator. Report slips and other falls separately, counts, misses, and binomial confidence intervals. This small exploratory sample does not support deployment claims. Ten 30 s non-fall episodes provide at most 300 s of exposure (0.0833 h); report actual valid duration and alarms separately for stable and recoverable conditions. Zero observed alarms over this exposure does not establish a low operational rate.

For matched events report $t_{alarm} - t_{onset}$ and $t_{alarm} - t_{contact}$ separately, including median, interquartile range, range, and count. Negative contact-relative delay denotes a pre-contact alarm. Report misses separately, and disclose buffering/filter delay. Simulator-time delay is not ESP32 inference latency or BLE delivery time.

B.6 Event-pipeline interoperability and completion

A future replay adapter shall emit existing v1 fields: `event_id`, `device_id`, `firmware_version`, uint32 `timestamp_ms`, confidence $P(\text{FALL})$, and signed per-channel maximum-magnitude peaks in g and degrees/s. Identify synthetic devices explicitly. Keep episode, simulator, model, and domain provenance in a linked sidecar. v1 does not carry the raw window; `subject_type=bipedal_robot` is a proposed extension, not a validated v1

routing feature.

Future fixtures must verify parser, storage, and report interoperability against the unchanged event contract. Those checks do not validate physical BLE or robot-specific explanations. Completion requires reproducible motion, calibrated timing/units, forty accepted episodes, a frozen split and annotations, recovered model preprocessing, held-out event/exposure metrics, and passing replay checks.

Human-model transfer remains exploratory. Report zero-shot and adapted performance separately; different body geometry, placements, controllers, and synthetic sensors may defeat transfer. Physical robot sensing, disturbance evaluation, onboard timing, mitigation control, and safety validation remain future work.